

CBEH Final Mock Examination

(Simulation 2)

Time Allowed: 90 Minutes

Total Questions: 70 (30 Cell Biology, 24 Histology, 12 Embryology, 4 Interdisciplinary)

Passing Requirement: Minimum 60% overall, with at least 50% in each module.

PART I: Cell Biology (30 Questions)

1. Multiple Choice: According to the fluid-mosaic model of the cell membrane, which of the following best describes the role of cholesterol?

- A. It acts as a primary active transporter for sodium and potassium ions.
- B. It acts as a fluidity buffer, preventing the membrane from becoming too stiff at low temperatures and too fluid at high temperatures.
- C. It forms the hydrophilic pores of gap junctions.
- D. It covalently links peripheral membrane proteins to the lipid bilayer.
- E. It is the primary ligand-binding site for G-protein coupled receptors.

2. True or False: Enveloped viruses acquire their lipid bilayer by budding through the host cell's plasma membrane or intracellular membranes, making them generally more susceptible to environmental degradation than naked viruses.

3. Open Question (Max 200 words): The CRISPR-Cas9 system is widely used for gene editing in medicine. Briefly explain the original, natural biological role of the CRISPR-Cas system in bacteria.

4. Fill in the gap: During DNA replication, the lagging strand is synthesized discontinuously in short segments known as _____ fragments, which are later joined together by DNA ligase.

5. Matching: Match the type of cell surface receptor with its functional mechanism.

- 1. G-protein coupled receptor (GPCR)
- 2. Receptor Tyrosine Kinase (RTK)
- 3. Ion channel-coupled receptor
- 4. Nuclear receptor

A. Opens to allow the flow of specific ions across the membrane upon ligand binding.

- B. Acts as a ligand-activated transcription factor inside the cell.*
- C. Activates an intracellular cascade by promoting GDP/GTP exchange on a heterotrimeric protein.*
- D. Dimerizes and auto-phosphorylates intracellular domains upon ligand binding.*

6. Multiple Choice: Which cytoskeletal filaments are characterized by dynamic instability, utilize GTP for polymerization, and serve as tracks for kinesin and dynein motor proteins?

- A. Microtubules
- B. Actin microfilaments
- C. Nuclear lamins
- D. Intermediate filaments
- E. Keratin filaments

7. True or False: Prion diseases, such as Creutzfeldt-Jakob disease, are caused by the transmission of misfolded proteins that induce normal versions of the same protein to adopt a pathological, aggregation-prone beta-sheet conformation.

8. Multiple Choice: During eukaryotic RNA processing, the spliceosome catalyzes the removal of introns. What are the primary functional components of the spliceosome?

- A. Transfer RNAs (tRNAs) and aminoacyl-tRNA synthetases
- B. Small nuclear ribonucleoproteins (snRNPs)
- C. DNA polymerases and helicases
- D. Ribosomal RNAs (rRNAs) and large ribosomal proteins
- E. RNA-induced silencing complexes (RISC)

9. Fill in the gap: Progression through the cell cycle is driven by the cyclic activation and degradation of regulatory proteins known as _____, which bind to and activate specific cyclin-dependent kinases (CDKs).

10. Multiple Choice: Which of the following mutations describes a single base substitution that changes a codon for an amino acid into a premature STOP codon?

- A. Missense mutation
- B. Silent mutation
- C. Frameshift mutation
- D. Nonsense mutation
- E. Splice-site mutation

11. Multiple Choice: A genetic condition is caused by a single gene mutation, but affected individuals exhibit diverse symptoms involving the eyes, cardiovascular system, and skeletal system. This phenomenon is an example of:

- A. Incomplete penetrance
- B. Polygenic inheritance
- C. Codominance

- D. Pleiotropy
- E. X-linked recessive inheritance

12. True or False: The primary structure of a protein is stabilized by non-covalent hydrogen bonds between the amino and carboxyl groups of the polypeptide backbone.

13. Multiple Choice: In the intrinsic pathway of apoptosis, the release of cytochrome c from mitochondria into the cytosol directly triggers the formation of which multiprotein complex?

- A. The membrane attack complex (MAC)
- B. The apoptosome
- C. The spliceosome
- D. The proteasome
- E. The kinetochore

14. Open Question (Max 200 words): Briefly define the concepts of stem cell potency. In your answer, differentiate between the developmental potential of embryonic stem cells (ESCs) and adult somatic stem cells.

15. Fill in the gap: The cytogenetic technique that utilizes fluorescently labeled DNA probes to detect specific DNA sequences on intact chromosomes is abbreviated as _____.

16. Multiple Choice: According to the “two-hit hypothesis” of cancer progression, which of the following genes generally requires loss-of-function mutations in both alleles to drive tumorigenesis?

- A. Proto-oncogenes
- B. Tumor suppressor genes
- C. Viral integrase genes
- D. Anti-apoptotic genes like Bcl-2
- E. Telomerase reverse transcriptase (TERT)

17. Multiple Choice: Which of the following best describes heterochromatin?

- A. Loosely packed chromatin containing actively transcribed genes.
- B. Chromatin that completely lacks histone proteins.
- C. Highly condensed chromatin that is typically transcriptionally inactive.
- D. Chromatin found exclusively in prokaryotic nucleoids.
- E. Chromatin that is only visible during the metaphase of mitosis.

18. Matching: Match the cellular organelle with its primary function.

1. Golgi apparatus
2. Peroxisome
3. Smooth Endoplasmic Reticulum

4. Lysosome

- A. *Contains acid hydrolases for the degradation of macromolecules.*
- B. *Synthesizes lipids and steroid hormones, and detoxifies drugs.*
- C. *Sorts, modifies, and packages proteins for targeted delivery.*
- D. *Performs lipid oxidation and detoxifies hydrogen peroxide.*

19. True or False: The wobble hypothesis explains why a single tRNA can recognize multiple codons, allowing the cell to translate all 61 amino acid codons with fewer than 61 distinct tRNA molecules.

20. Multiple Choice: The two strands of the DNA double helix are said to be “antiparallel.” This means that:

- A. One strand is made of RNA and the other is DNA.
- B. One strand is composed of purines and the other of pyrimidines.
- C. The 5' to 3' direction of one strand runs opposite to the 5' to 3' direction of the other.
- D. They contain identical nucleotide sequences.
- E. One strand replicates conservatively, while the other replicates semi-conservatively.

21. Fill in the gap: The highly specific protein complex responsible for degrading ubiquitinated, misfolded proteins in the cytosol is the _____.

22. Multiple Choice: In eukaryotic transcription, the TATA box is a critical DNA sequence located within the:

- A. Terminator region
- B. Intron
- C. Enhancer element
- D. Promoter region
- E. Ribosomal binding site

23. Multiple Choice: Homologous recombination (crossing-over) is a key mechanism generating genetic diversity. During which specific meiotic stage does crossing-over occur?

- A. Diakinesis
- B. Pachytene of Prophase I
- C. Metaphase II
- D. Anaphase I
- E. Zygotene of Prophase I

24. True or False: During prometaphase of mitosis, the nuclear envelope breaks down, allowing spindle microtubules to attach to the kinetochores of chromosomes.

25. Multiple Choice: Which of the following molecules functions as a second messenger in the phospholipase C (PLC) signaling pathway?

- A. Cyclic AMP (cAMP)
- B. Protein Kinase A (PKA)
- C. Inositol 1,4,5-trisphosphate (IP₃)
- D. GTP
- E. ATP

26. Matching: Categorize the following features as characteristic of Prokaryotes, Eukaryotes, or Both.

1. Circular DNA suspended in a nucleoid region
2. Membrane-bound organelles (e.g., mitochondria)
3. Ribosomes used for protein synthesis
4. Intron splicing of pre-mRNA

A. Prokaryotes only

B. Eukaryotes only

C. Both Prokaryotes and Eukaryotes

27. Fill in the gap: During receptor-mediated endocytosis, the cytosolic surface of the invaginating plasma membrane is coated with a lattice-forming protein called _____.

28. Open Question (Max 200 words): Normal cells grown in tissue culture stop dividing when they form a confluent monolayer. Explain the concept of “in vitro contact inhibition” and describe how its loss reflects the altered cell behavior in cancer and tumor progression in vivo.

29. Multiple Choice: To integrate into the host cell's genome, retroviruses like HIV must synthesize double-stranded DNA from their RNA genome. Which viral enzyme performs this function?

- A. RNA polymerase II
- B. Caspase-3
- C. Integrase
- D. Reverse transcriptase
- E. Telomerase

30. Multiple Choice: The tumor suppressor protein p53 is famously known as the “guardian of the genome.” Its primary role at the G₁/S cell cycle checkpoint is to:

- A. Phosphorylate the Retinoblastoma (Rb) protein to stimulate S phase entry.
 - B. Initiate the degradation of securin to trigger anaphase.
 - C. Halt the cell cycle in response to DNA damage to allow for repair or initiate apoptosis.
 - D. Recruit DNA helicase to the origins of replication.
 - E. Act as a primary growth factor receptor on the plasma membrane.
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PART II: Histology (24 Questions)

31. Multiple Choice: A lining epithelium that must accommodate significant stretching and volume changes, such as that found in the urinary bladder, is classified as:

- A. Stratified squamous keratinized epithelium
- B. Simple columnar epithelium
- C. Transitional epithelium (Urothelium)
- D. Pseudostratified ciliated epithelium
- E. Simple cuboidal epithelium

32. True or False: Microvilli are motile apical surface specializations composed of a core of 9+2 microtubules, whereas cilia are non-motile structures supported by actin filaments.

33. Open Question (Max 200 words): Describe the distinct functional roles of osteoblasts, osteocytes, and osteoclasts in the processes of bone formation and bone remodeling.

34. Fill in the gap: Exocrine glands that release their secretory product through the complete rupture and death of the entire cell (e.g., sebaceous glands) utilize a _____ mechanism of secretion.

35. Multiple Choice: Which resident cell of connective tissue proper is characterized by abundant basophilic cytoplasmic granules containing histamine and heparin, playing a major role in allergic and inflammatory responses?

- A. Plasma cell
- B. Fibroblast
- C. Adipocyte
- D. Macrophage
- E. Mast cell

36. Matching: Match the mature blood cell with its primary histological function or feature.

- 1. Erythrocyte
- 2. Neutrophil
- 3. Eosinophil
- 4. Monocyte

A. Primary phagocyte of bacteria; features a multi-lobed nucleus.

B. Anucleate biconcave disc specialized for oxygen transport.

C. Large agranulocyte with a kidney-shaped nucleus; differentiates into tissue macrophages.

D. Granulocyte involved in combating parasitic infections and modulating allergic reactions.

37. Multiple Choice: A histological section of a primary lymphoid organ reveals a lobulated structure with a dark outer cortex and a lighter inner medulla. The medulla contains distinctive concentric whorls of epithelial reticular cells. Which organ is this?

- A. Spleen
- B. Palatine tonsil
- C. Lymph node
- D. Thymus
- E. Bone marrow

38. True or False: Smooth muscle tissue is characterized by the absence of visible cross-striations because its actin and myosin myofilaments are not arranged in highly ordered sarcomeres.

39. Multiple Choice: In myelinated axons of the peripheral nervous system, saltatory conduction occurs. This rapid transmission is facilitated by gaps in the myelin sheath known as:

- A. Synaptic clefts
- B. Nodes of Ranvier
- C. Schmidt-Lanterman clefts
- D. Motor endplates
- E. Haversian canals

40. Fill in the gap: The most abundant type of cartilage in the human body, which provides a template for endochondral ossification and contains primarily type II collagen, is _____ cartilage.

41. Multiple Choice: In the extracellular matrix (ECM) of connective tissue, which of the following is a massive, non-sulfated glycosaminoglycan (GAG) that provides the tissue with the ability to resist compressive forces by retaining water?

- A. Fibronectin
- B. Type I Collagen
- C. Hyaluronic acid
- D. Elastin
- E. Laminin

42. Multiple Choice: In skeletal muscle histology, a “triad” refers to:

- A. The association of one T-tubule and two terminal cisternae of the sarcoplasmic reticulum.
- B. Three parallel myosin thick filaments.
- C. A motor neuron, a neuromuscular junction, and a muscle fiber.

- D. The epimysium, perimysium, and endomysium connective tissue layers.
- E. The arrangement of an A-band, I-band, and Z-disc.

43. Matching: Match the neuroglial cell with its corresponding function.

- 1. Astrocyte
- 2. Oligodendrocyte
- 3. Schwann cell
- 4. Ependymal cell

A. Myelinates multiple axons within the Central Nervous System.

B. Forms the epithelial-like lining of the brain ventricles and produces cerebrospinal fluid.

C. Myelinates a single internode of an axon in the Peripheral Nervous System.

D. Provides structural/metabolic support to CNS neurons and helps maintain the blood-brain barrier.

44. True or False: Endochondral ossification, the process by which long bones grow, requires a pre-existing hyaline cartilage template.

45. Fill in the gap: Intercellular communication between adjacent cardiac muscle cells is achieved through specialized channels called _____ junctions, which are formed by hexameric protein complexes known as connexons.

46. Multiple Choice: In the epidermis of thick skin, which layer contains the actively dividing stem cells that continually regenerate the epithelium?

- A. Stratum corneum
- B. Stratum granulosum
- C. Stratum lucidum
- D. Stratum basale (germinativum)
- E. Stratum spinosum

47. Multiple Choice: In the spleen, the primary functional distinction between the white pulp and the red pulp is that:

- A. White pulp filters blood to remove old erythrocytes, while red pulp produces new erythrocytes.
- B. White pulp produces red blood cells, while red pulp stores white blood cells.
- C. White pulp is composed of lymphoid tissue for immune responses, while red pulp filters the blood to remove senescent erythrocytes.
- D. White pulp is exclusively found in the thymus, while red pulp is found in the spleen.
- E. White pulp contains osteoblasts, while red pulp contains hematopoietic stem cells.

48. Open Question (Max 200 words): A peripheral nerve is structurally organized and protected by three distinct layers of connective tissue. Describe the structural

arrangement of the endoneurium, perineurium, and epineurium.

49. True or False: Platelets (thrombocytes) are true cells with lobulated nuclei that differentiate directly from myeloid stem cells in the bloodstream.

50. Multiple Choice: In histochemistry, the Periodic Acid-Schiff (PAS) stain is primarily utilized to visualize which of the following cellular components?

- A. Nucleic acids (DNA/RNA)
- B. Neutral lipids and triglycerides
- C. Calcium deposits
- D. Carbohydrates (such as glycogen and mucins)
- E. Mitochondria

51. Fill in the gap: The endocrine gland characterized by spherical functional units called follicles, which store a gelatinous colloid used for hormone synthesis, is the _____ gland.

52. Multiple Choice: Histologically, cardiac muscle can be distinguished from skeletal muscle by the presence of:

- A. Multiple, peripheral nuclei per cell.
- B. A complete lack of cross-striations.
- C. Branching fibers and intercalated discs.
- D. Motor endplates for voluntary control.
- E. Dense bodies instead of Z-discs.

53. Multiple Choice: Which specialized connective tissue forms the framework of the epiglottis and the external ear (pinna)?

- A. Fibrocartilage
- B. Dense regular connective tissue
- C. Elastic cartilage
- D. Hyaline cartilage
- E. Spongy bone

54. Multiple Choice: Which instrument/method uses laser light to illuminate a single focal plane while rejecting out-of-focus light to generate high-resolution, 3D optical sections of a biological specimen?

- A. Transmission Electron Microscopy (TEM)
 - B. Scanning Electron Microscopy (SEM)
 - C. Bright-field light microscopy
 - D. Confocal fluorescence microscopy
 - E. Flow cytometry
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PART III: Embryology (12 Questions)

55. Multiple Choice: During the ovarian cycle, ovulation is directly triggered by a sudden surge in which hormone?

- A. Progesterone
- B. Estrogen
- C. Luteinizing Hormone (LH)
- D. Human Chorionic Gonadotropin (hCG)
- E. Follicle-Stimulating Hormone (FSH)

56. True or False: In the seminiferous tubules, the blood-testis barrier is formed by tight junctions between adjacent Leydig cells, protecting developing sperm from autoimmune attack.

57. Fill in the gap: About 3 to 4 days after fertilization, the dividing zygote forms a solid ball of 16-32 cells known as a _____, which shortly after develops a fluid-filled cavity to become a blastocyst.

58. Open Question (Max 200 words): Gastrulation occurs during the 3rd week of human development. Describe the role of the primitive streak and primitive node in the formation of the three primary germ layers.

59. Multiple Choice: During early implantation, the trophoblast layer of the blastocyst differentiates into two distinct layers. The outer, multinucleated layer that aggressively invades the maternal endometrium is the:

- A. Cytotrophoblast
- B. Syncytiotrophoblast
- C. Epiblast
- D. Hypoblast
- E. Extraembryonic mesoderm

60. Matching: Match the developing nervous system structure with its embryonic derivative.

1. Central Nervous System (Brain and Spinal Cord)
2. Peripheral Nervous System (e.g., dorsal root ganglia)
3. Nucleus pulposus of the intervertebral disc
4. Epithelial lining of the digestive tract

- A. Endoderm
- B. Neural Tube
- C. Neural Crest Cells
- D. Notochord

61. True or False: In a healthy human pregnancy, maternal and fetal blood mix directly within the intervillous spaces of the placenta to ensure optimal nutrient and gas exchange.

62. Fill in the gap: During the fourth week of development, the lateral and craniocaudal folding of the embryo converts the flat trilaminar disc into a cylinder, incorporating the dorsal part of the yolk sac to form the primitive _____ tube.

63. Multiple Choice: The notochord is a critical signaling structure that induces the formation of the neural plate. From which primary germ layer does the notochord arise?

- A. Ectoderm
- B. Endoderm
- C. Mesoderm
- D. Extraembryonic mesoderm
- E. Trophoblast

64. Multiple Choice: Congenital omphalocele and gastroschisis are developmental defects related to the failure of the abdominal wall to close properly around which developing organ system?

- A. Respiratory system
- B. Cardiovascular system
- C. Urogenital system
- D. Central nervous system
- E. Digestive system (Midgut)

65. Multiple Choice: Which of the following best defines the process of **spermiogenesis**?

- A. The mitotic proliferation of spermatogonia.
- B. The reductional division of primary spermatocytes into secondary spermatocytes.
- C. The entire sequence of events from spermatogonium to mature spermatozoon.
- D. The structural and functional maturation of round haploid spermatids into highly specialized, motile spermatozoa.
- E. The capacitation of sperm within the female reproductive tract.

66. Multiple Choice: Prior to gastrulation, the inner cell mass (embryoblast) organizes into a bilaminar embryonic disc composed of:

- A. Ectoderm and Endoderm
 - B. Syncytiotrophoblast and Cytotrophoblast
 - C. Epiblast and Hypoblast
 - D. Somatic and Splanchnic mesoderm
 - E. Amnion and Yolk sac
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PART IV: Interdisciplinary (4 Questions)

67. Matching: *[Cell Biology + Histology + Embryology]* Match the clinical disorder/developmental defect with its primary molecular or cellular etiology.

1. Epidermolysis bullosa
 2. Kartagener syndrome (Primary Ciliary Dyskinesia)
 3. Lysosomal storage diseases (e.g., Tay-Sachs)
 4. Osteoporosis
- A. Defect in microtubule motor proteins (dynein) causing immotile cilia; leads to respiratory issues and situs inversus.*
- B. Imbalance in bone remodeling where osteoclast resorption outpaces osteoblast deposition.*
- C. Defects in intermediate filaments (keratin) or hemidesmosomes causing fragile epithelial-connective tissue junctions.*
- D. Genetic deficiency of acid hydrolase enzymes, leading to the toxic accumulation of undegraded macromolecules.*

68. Open Question (Max 200 words): *[Histology + Embryology]* During the 4th week of development, neural crest cells undergo an epithelial-to-mesenchymal transition and migrate extensively throughout the body. Briefly describe the developmental origin of neural crest cells. Then, list two specific histological structures or cell types they give rise to within the peripheral nervous system (PNS).

69. Open Question (Max 200 words): *[Cell Biology + Histology]* Epidermolysis bullosa is a severe genetic disease causing skin blistering and impaired tissue integrity. Connect the cell biology of the epithelial cytoskeleton (specifically intermediate filaments) to the histology of the anchoring junctions (hemidesmosomes) that bind the epidermis to the underlying basement membrane.

70. Multiple Choice: *[Cell Biology + Embryology]* Concerning mitochondrial inheritance and cellular energy during human reproduction, which of the following statements is correct?

- A. Mitochondria in early embryos are inherited purely from the sperm, as oocytes lack functional mitochondria.
- B. The mitochondrial matrix is topologically equivalent to the lumen of the endoplasmic reticulum.
- C. Mitochondrial transport along developing axons relies exclusively on actin microfilaments.
- D. Early embryonic cleavage heavily relies on maternally inherited mitochondria to supply ATP for rapid cell divisions.

E. Cytochrome c is actively imported into the nucleus to drive the initial stages of fertilization.

ANSWER KEY

Part I: Cell Biology

1. **B**
2. **True**
3. **Open Concept:** In bacteria, the CRISPR-Cas system acts as an adaptive immune defense mechanism against bacteriophages (viruses). Short segments of viral DNA are integrated into the bacterial CRISPR locus. These sequences are transcribed into guide RNAs, which direct the Cas endonuclease to recognize and cleave complementary sequences upon viral re-infection.
4. **Okazaki**
5. **Matching:** 1-C, 2-D, 3-A, 4-B
6. **A**
7. **True**
8. **B**
9. **Cyclins**
10. **D**
11. **D**
12. **False** (Primary structure is stabilized by strong, covalent peptide bonds; hydrogen bonds stabilize secondary structures like alpha helices and beta sheets).
13. **B**
14. **Open Concept:** Stem cell potency refers to the differentiation potential of a cell. Embryonic stem cells (ESCs) derived from the blastocyst inner cell mass are *pluripotent*, meaning they can form all three embryonic germ layers and any somatic cell type in the body. Adult somatic stem cells are generally *multipotent*, meaning their developmental potential is restricted to forming the specific cell

lineages of their resident tissue (e.g., hematopoietic stem cells only form blood cells).

15. **FISH**

16. **B**

17. **C**

18. **Matching:** 1-C, 2-D, 3-B, 4-A

19. **True**

20. **C**

21. **Proteasome**

22. **D**

23. **B**

24. **True**

25. **C**

26. **Matching:** 1-A (Prokaryote), 2-B (Eukaryote), 3-C (Both), 4-B (Eukaryote)

27. **Clathrin**

28. **Open Concept:** Normal somatic cells exhibit contact inhibition in vitro; when they touch neighboring cells, mechanical and chemical signaling pathways halt the cell cycle, preventing overgrowth. Cancer cells have mutations (e.g., in tumor suppressors or oncogenes) that disable this sensing mechanism. Loss of contact inhibition in vivo allows cancer cells to continuously proliferate, piling up into multi-layered solid tumors and subsequently invading surrounding tissues.

29. **D**

30. **C**

Part II: Histology

31. **C**

32. **False** (Cilia are composed of motile 9+2 microtubules; microvilli are non-motile surface extensions supported by an actin core).

33. **Open Concept:** Osteoblasts are bone-forming cells that secrete the organic extracellular matrix (osteoid). When osteoblasts become trapped within the calcified matrix in lacunae, they differentiate into osteocytes, which act as mechanosensors to maintain the bone tissue. Osteoclasts are large, multinucleated macrophages that resorb (break down) bone matrix by secreting acid and enzymes, allowing for continuous bone remodeling and calcium release.

34. **Holocrine**

35. **E**

36. **Matching:** 1-B, 2-A, 3-D, 4-C

37. **D**

38. **True**

39. **B**

40. **Hyaline**

41. **C**

42. **A**

43. **Matching:** 1-D, 2-A, 3-C, 4-B

44. **True**

45. **Gap**

46. **D**

47. **C**

48. **Open Concept:** The endoneurium is a delicate loose connective tissue layer surrounding individual myelinated or unmyelinated axons. The perineurium is a specialized, dense cellular connective tissue layer that bundles multiple axons into fascicles and forms the blood-nerve barrier. The epineurium is the outermost dense irregular connective tissue layer binding multiple fascicles together to form the entire peripheral nerve.

49. **False** (Platelets are not true cells; they are small, anucleate cytoplasmic fragments derived from large bone marrow cells called megakaryocytes).

50. **D**

51. **Thyroid**

52. **C**

53. **C**

54. **D**

Part III: Embryology

55. **C**

56. **False** (The blood-testis barrier is formed by tight junctions between Sertoli cells, not Leydig cells).

57. **Morula**

58. **Open Concept:** The primitive streak forms on the epiblast surface, establishing the embryo's craniocaudal axis. Epiblast cells migrate toward the streak, detach, and invaginate (gastrulation). Some displace the hypoblast to form embryonic endoderm; others migrate between the epiblast and endoderm to form intraembryonic mesoderm. Cells migrating specifically through the primitive node form the notochord. The remaining epiblast cells become the embryonic ectoderm.

59. **B**

60. **Matching:** 1-B, 2-C, 3-D, 4-A

61. **False** (Maternal blood bathes the trophoblast villi, but it is physically separated from fetal blood by the placental barrier layers; they do not mix directly).

62. **Gut** (or digestive/intestinal)

63. **C**

64. **E**

65. **D**

66. **C**

Part IV: Interdisciplinary

67. **Matching:** 1-C, 2-A, 3-D, 4-B

68. **Open Concept:** Neural crest cells originate from the dorsal margins (crests) of the neural folds during neurulation (ectoderm lineage). As the neural tube closes, they undergo an epithelial-to-mesenchymal transition and migrate away. In the

peripheral nervous system, they form sensory (dorsal root) ganglia, autonomic (sympathetic/parasympathetic) ganglia, and peripheral glial cells, such as Schwann cells and satellite cells.

69. **Open Concept:** Epidermolysis bullosa involves genetic defects in proteins anchoring the epidermis to the dermis. Inside the basal epithelial cells, the cytoskeleton relies on keratin intermediate filaments for tensile strength. These filaments insert into hemidesmosomes at the basal domain. If keratin or hemidesmosome components (like integrins or laminins) are mutated, the mechanical link fails, causing the epithelial layer to shear off from the underlying basement membrane upon slight friction, producing severe blisters.

70. **D**